Exp. No: 2

Electric Field and Electric Flux Density in a Sphere Full of Charges with Varying Radius

Aim:

To write and validate MATLAB programs for electric field intensity (E) and electric flux density (D) for specified spherical charge distributions.

Objectives:

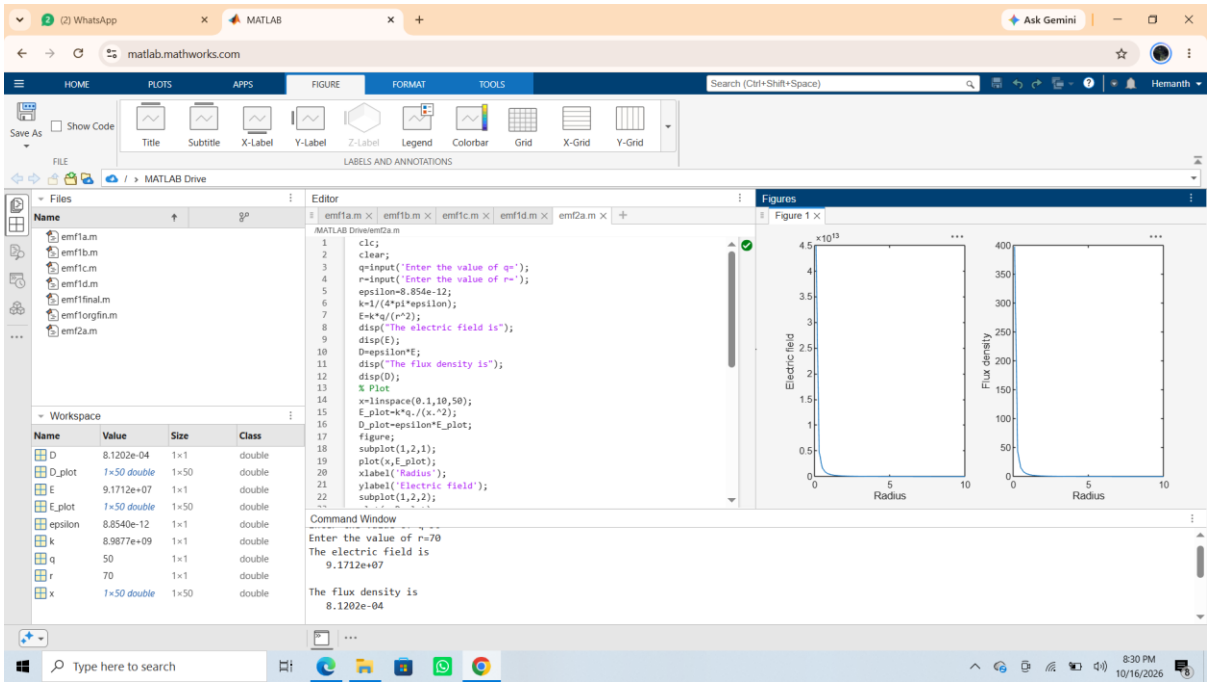
1. To derive expressions for electric field intensity (E) and electric flux density (D) inside and outside a uniformly charged sphere using Gauss's law.
2. To plot the variation of E and D with radial distance for varying sphere radius and charge density.
3. To validate the simulated results against theoretical calculations for different radii.

Test case (i) Determination and plotting of electric field and electric flux density in sphere full of charges with varying radius

```
clc;
clear; q=input('Enter the value of q=');
r=input('Enter the value of r=');
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
E=k*q/(r^2);
disp("The electric field is");
disp(E);
D=epsilon*E;
disp("The flux density is");
disp(D);
% Plot
x=linspace(0.1,10,50);
```

```
E_plot=k*q./(x.^2);
D_plot=epsilon*E_plot;
figure;
subplot(1,2,1);
plot(x,E_plot);
xlabel('Radius');
ylabel('Electric field');
subplot(1,2,2);
plot(x,D_plot);
xlabel('Radius');
ylabel('Flux density')
```

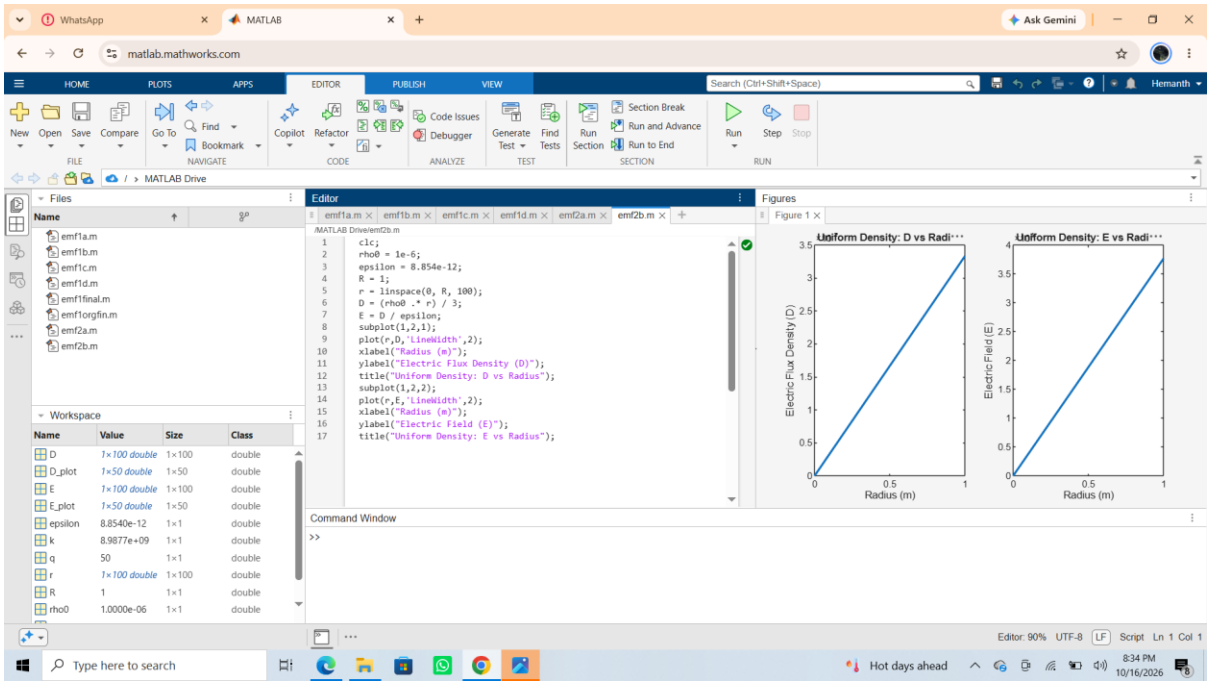
OUTPUT:



Test case (ii) Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0

Program:

```
clc;
rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;
r = linspace(0, R, 100);
D = (rho0 .* r) / 3;
E = D / epsilon;
subplot(1,2,1);
plot(r,D,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Flux Density (D)");
title("Uniform Density: D vs Radius");
subplot(1,2,2);
plot(r,E,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Field (E)");
title("Uniform Density: E vs Radius);
;
```



Test case (iii) For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive and plot the electric field (E) and electric flux density (D)

Program:

```
clc;

clear;

close all;


% Given data

rho0 = 1e-6;

epsilon = 8.854e-12;

R = 1;


% Radius vector

r = linspace(0, R, 100);

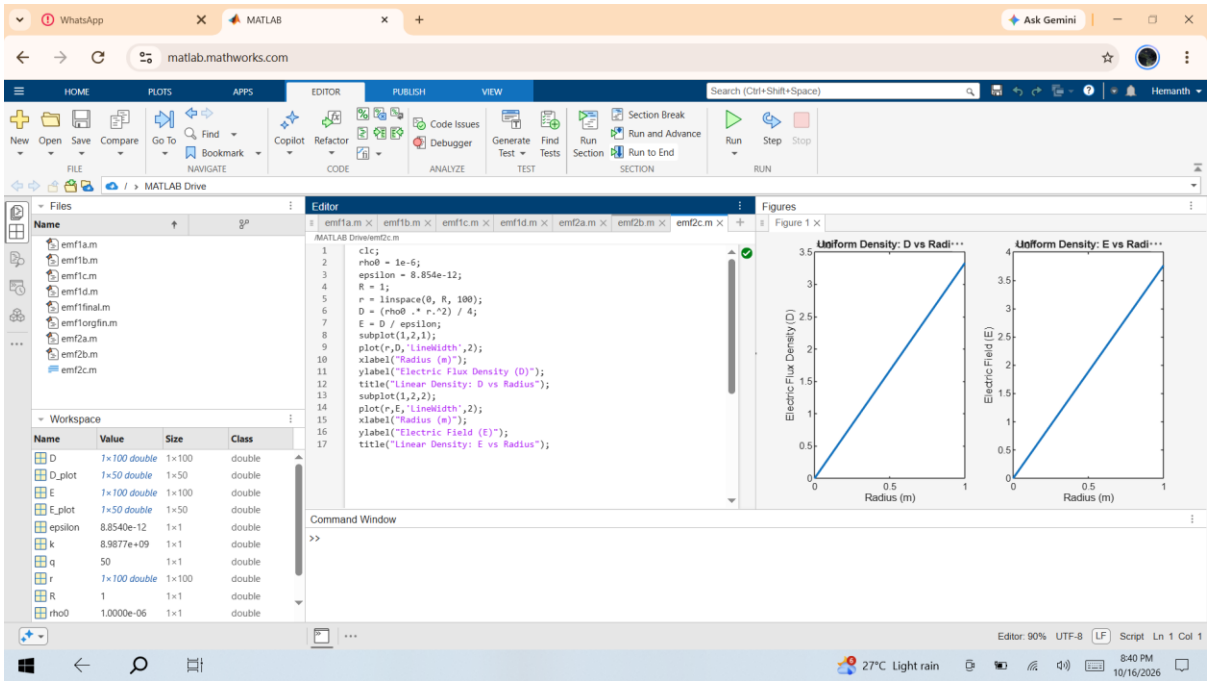

% Electric flux density (from Gauss's law)

D = (rho0 .* r.^2) / 4;
```

```
% Electric field
E = D / epsilon;

% Plot D vs r
subplot(1,2,1);
plot(r, D, 'LineWidth', 2);
xlabel('Radius (m)');
ylabel('Electric Flux Density D (C/m^2)');
title('Linear Density: D vs Radius');
grid on;

% Plot E vs r
subplot(1,2,2);
plot(r, E, 'LineWidth', 2);
xlabel('Radius (m)');
ylabel('Electric Field E (V/m)');
title('Linear Density: E vs Radius');
grid on;
```



Program:

Test case (iv): For a uniformly charged solid sphere of radius R = 2 m and volume charge density $\rho_v = 3\times10^{-6}$ C/m^3, calculate and plot E and D inside and outside the sphere.

```
clc; clear; close all;

 rho0 = 3e-6;          % Uniform volume charge density (C/m^3)

epsilon0 = 8.854187817e-12;      % Permittivity of free space (F/m)

 R = 2;                  % Sphere radius (m)

 r = linspace(0,4,401);       % Radial distance (m)

 E = zeros(size(r));

 inside = (r <= R);

 outside = (r > R);

 E(inside) = rho0 .* r(inside) ./ (3epsilon0);

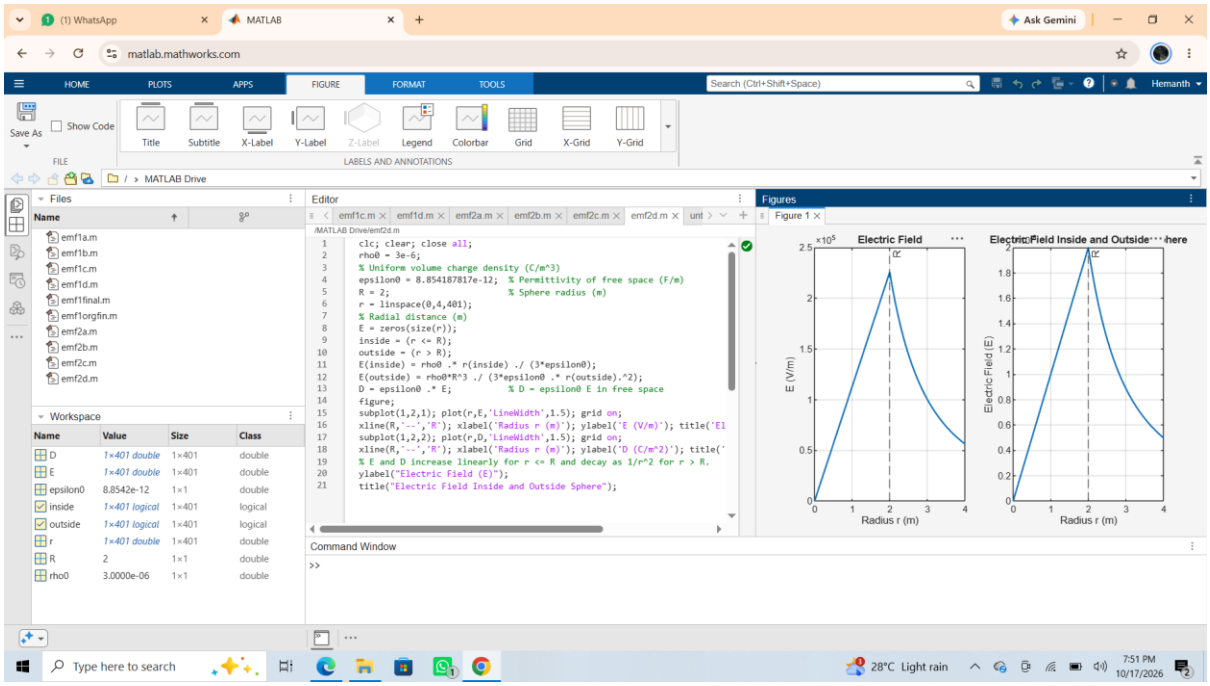
 E(outside) = rho0R^3 ./ (3epsilon0 . r(outside).^2);

 D = epsilon0 .* E;                    % D = epsilon0 E in free space

 figure;
```

```
subplot(1,2,1); plot(r,E,'LineWidth',1.5); grid on;
xline(R,'--','R'); xlabel('Radius r (m)'); ylabel('E (V/m)'); title('Electric Field');
subplot(1,2,2); plot(r,D,'LineWidth',1.5); grid on;
xline(R,'--','R'); xlabel('Radius r (m)'); ylabel('D (C/m^2)'); title('Electric Flux Density');
% E and D increase linearly for r <= R and decay as 1/r^2 for r > R.
ylabel("Electric Field (E)");
title("Electric Field Inside and Outside Sphere");
```

OUTPUT:



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained

1	Select the appropriat e charge distributi o n model	Select the appropriat e charge distributio n model	Correctly distinguishe s point, uniform volume and specified non uniform charge models and their assumptions .	Selects the correct model with minor clarific a tion needed	Shows partial understandin g of the charge model.	Uses an unsuitable model or assumptions.	25	2.5	
2	Impleme nt E and D calculatio n s in MATLA B	Implement E and D calculation s in MATLAB	Correctly calculates electric field E and flux density D with consistent SI units.	Calcula t ions are mostly correct with minor coding or unit errors..	Some calculations work but require assistance..	Unable to obtain valid E and D values.	25	2.5	
3	Analyze radial variation and validate result	Analyze radial variation and validate results	Plots the correct inside/outsid e or stated distribution behaviour and verifies against theory.	Plots and validati on are mostly correct with minor discrep a ncies.	Plots or theoretical validation are incomplete.	Plots are incorrect or results are not validated.	30	3	
4	Interpret and document findings	Interpret and document findings	Explains the effect of radius and charge distribution and presents labelled plots/tables.	Interpre t ation is general l y correct with minor omissio ns.	Interpretatio n is limited or documentati o n incomplete..	Cannot interpret or document results.	20	2	
Tota l							100	10	